

LAVANYA MITTAL
Panchkula, Haryana, India
Phone: 8570826991 | Email: lavanyamittal07@gmail.com
LinkedIn: www.linkedin.com/in/lavanyamittal
GitHub: <https://github.com/Lavanya2104>

PROFESSIONAL SUMMARY
Passionate mechatronics engineer with a strong focus on robotics, automation, and control systems. Skilled in integrating mechanical design with embedded systems and software, leveraging expertise in ROS, machine vision, and sensor fusion

INTERNSHIP EXPERIENCE	
COMPANY NAME	Autoliv India
DESIGNATION	Process Engineering Intern (Industrial Engineering)
DURATION	Feb 2025 – Jan 2026
KEY CONTRIBUTIONS	- Led Line Balancing Activities Across Multiple Airbag Production Lines, Contributing to a ~20% Improvement in Productivity. - Represented The Team at The ACMA 22nd Regional Quality Circle Competition, Securing 3rd Place in The Sub-Category. - Provided Engineering Support for Both New Projects and Serial Production Lines, Ensuring Process Readiness and Compliance. - Actively Participated in Customer Audits and Approval Processes, Supporting Documentation, Process Validation, And Poka – Yoke Implementations. - Developed A Jetson Orin Nano–Based Computer Vision Proof-Of-Concept for Steering Wheel Leather Wrapped Thread Colour Detection. - Worked Closely with Cross-Functional Teams to Support Process Standardization, Efficiency Improvements, And Safety-Critical Automotive Manufacturing.

PROJECT EXPERIENCE		
EMBEDDED & ROBOTICS PROJECTS		
PROJECT TITLE	NRF Remote Control	Color Sorting Robotic Arm with Computer Vision
DURATION	July 2023 – Oct 2023	June 2024 – Nov 2024
KEY CONTRIBUTIONS	Designed and tested an NRF-based wireless receiver module for reliable remote communication	Developed a Python-based colour detection system integrated with a robotic arm for automated sorting.
EMBEDDED & ROBOTICS PROJECTS		
PROJECT TITLE	Omni Directional Bot	Thread Colour Detection
DURATION	Feb 2026 – Ongoing	Feb 2025 – March 2025
KEY CONTRIBUTIONS	Designed, Assembled, And Programmed Mobile Robot with A Custom Bluetooth Control Application Developed Using MIT App Inventor	Implemented a real-time computer vision on NVIDIA Jetson Orin Nano for thread colour detection on Leather Wrapped Steering Wheel
ROBOTICS & CONTROL SYSTEMS PROJECTS		
PROJECT TITLE	TurtleBot (AMR Prototype)	BaJA SAEINDIA 2025 - ABaJA
DURATION	July 2024 – Jan 2025	Dec 2024 – Jan 2025
KEY CONTRIBUTIONS	Built an autonomous mobile robot using LiDAR and wheel encoders for perception and navigation.	Designed the software architecture for Adaptive Cruise Control (ACC) and contributed to system-level design documentation and Phase 1 technical presentation

TECHNICAL SKILLS	
SOFTWARE & PROGRAMMING	EMBEDDED & HARDWARE
- Python - C/C++ (Arduino) - ROS (Basics) - MATLAB (Basics)	- Arduino - Raspberry Pi - NVIDIA Jetson Orin Nano - NRF24L01 - Motor Drivers, Encoders
COMPUTER VISION & AI	ROBOTICS & CONTROL SYSTEMS
- OpenCV - Image Processing - Camera Integration - Real-time CV on Edge Devices	- LiDAR Integration - Autonomous Mobile Robots - Line Following Algorithms - ACC (Concept & Architecture)
MANUFACTURING & PROCESS ENGINEERING	
- Line Balancing - Operator Movement Analysis - Cycle Time Study - SWCT - Takt Time & Line Capacity Analysis - Process Documentation - New & Serial Production Support - Poka – Yoke Implementation - Automotive Quality & Safety Standards Adherence	

ACADEMIC QUALIFICATIONS				
Bachelors of Engineering	Mechatronics Engineering	Chitkara University	2022-2026	8.6 CGPA
Class XII	Non - Medical	CBSE	2021-2022	83.6 %
Class X		ICSE	2019-2020	94%

CERTIFICATIONS/COURSES
- Elite Design Workshop: Mastering PCB Layouts (5-Day Hands-On) — Feb 2024 - Online Course: Overview of Space Science, Technology & Applications — Mar 2023 - Workshop: Innovative Solutions for Modern Manufacturing (Laser Cutting) — Apr 2023

EXTRACURRICULAR / ACHIEVEMENTS
🏆 Secured 1st Position in Quality Quiz conducted at Autoliv India 🥈 Won 2 Silver Medals in Table Tennis at SCC (Sweden Corporate Championship) 🏆 All India 3rd Rank — BAJA SAEINDIA 2025 (Phase-1 Presentation) 🏆 3rd Place — ACMA 22nd Regional Quality Circle Competition (Sub – Category)

LANGUAGES
- English (Professional Working Proficiency) - Hindi (Native)